

STEP: A Framework for Solving Scientific Video Inverse Problems with Spatiotemporal Diffusion Priors

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Abstract

Reconstructing spatially and temporally coherent videos from time-varying measurements is a fundamental challenge in many scientific domains. A major difficulty arises from the sparsity of measurements, which hinders accurate recovery of temporal dynamics. Existing image diffusion-based methods rely on extracting temporal consistency directly from measurements, limiting their effectiveness on scientific tasks with high spatiotemporal uncertainty. We address this difficulty by proposing a plug-and-play framework that incorporates a learned spatiotemporal diffusion prior. Due to its plug-and-play nature, our framework can be flexibly applied to different video inverse problems without the need for task-specific design and temporal heuristics. We further demonstrate that a spatiotemporal diffusion model can be trained efficiently with limited video data. We validate our approach on two challenging scientific video reconstruction tasks—black hole video reconstruction and dynamic MRI. While baseline methods struggle to provide temporally coherent reconstructions, our approach achieves significantly improved recovery of the spatiotemporal structure of the underlying ground truth videos.

1 Introduction

Reconstructing high-quality videos from time-varying measurements is a core challenge in many scientific domains, such as black hole video reconstruction [21] and dynamic magnetic resonance imaging (MRI) [24], where recovered spatiotemporal features influence downstream scientific analysis or medical interpretation. These problems are inherently difficult due to the high dimensionality of the underlying video and the severe loss of both spatial and temporal information during acquisition.

Diffusion models (DMs) have become the state-of-the-art for modeling image and video distributions, offering powerful priors for solving inverse problems [78]. However, most existing diffusion-based approaches for video inverse problems are motivated by either video editing tasks or restoration problems in photography [15, 36, 35, 73], where the spatiotemporal structure is either given or mostly preserved in the measurements. These methods leverage image diffusion models (IDMs) to process each video frame independently while enforcing temporal consistency via correlated noise or optical flow. We find that these methods struggle on scientific problems where there is limited spatial structure in the measurements to capture complex dynamics.

Plug-and-play diffusion prior (PnPDP) methods offer a general strategy for solving inverse problems by leveraging pre-trained diffusion models as priors. They have shown strong performance across a range of applications, including natural image restoration [63, 50, 32, 11, 44, 39, 81], medical imaging [29, 49, 13, 28, 17, 12, 75], and physics-based inverse problems [78, 1, 23, 51, 52, 70]. Despite being compatible with videos in principle, existing PnPDP methods are predominantly investigated for static images. It remains largely unexplored how to effectively obtain a well-trained video diffusion prior and extend the PnPDP framework to ill-posed video inverse problems.

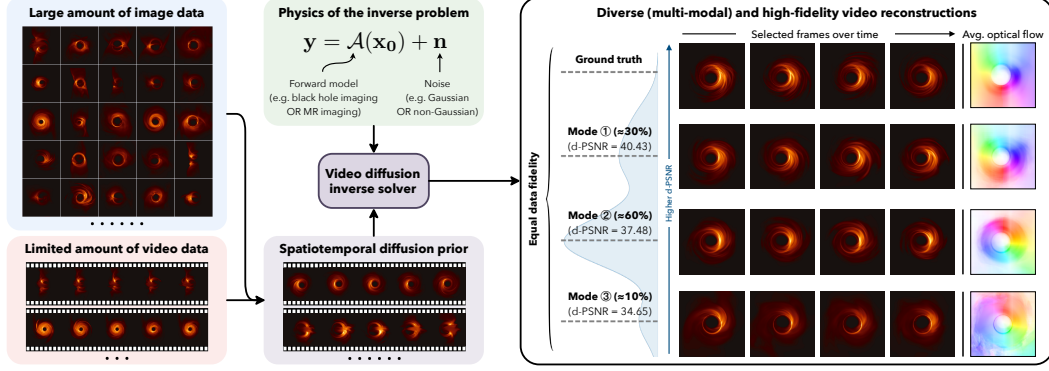


Figure 1: An overview of our proposed framework with Spatiotemporal Diffusion Priors, STEP, for scientific video inverse problems. Left: STEP combines the physic model of the target problem with a spatiotemporal diffusion prior that directly characterizes the video distribution. We show that such a prior can be efficiently obtained by fine-tuning a pre-trained image diffusion model with limited video data. Right: STEP can generate diverse solutions to a black hole video reconstruction problem that exhibit equally good fidelity with the measurements.

In this paper, we introduce STEP, a framework for solving scientific video inverse problems by integrating a **S**patio**T**emporal video diffusion **P**rior into a PnPDP method. STEP enforces learned priors over both spatial and temporal dimensions and does not rely on heuristics to ensure temporal consistency, making it well-suited for tasks with high spatiotemporal uncertainty. To obtain a spatiotemporal prior efficiently, we are inspired by [62] to first train an image latent diffusion model with a 2D UNet and then turn it into a spatiotemporal video diffusion model by adding a zero-initialized temporal module to each 2D convolution module in the UNet. This enables us to fine-tune a spatiotemporal diffusion prior from a pre-trained IDM in a data- and time-efficient manner, using only hundreds to thousands of videos and a few hours of training on a single A100 GPU. We then integrate the resulting prior with the Decoupled Annealing Posterior Sampling (DAPS) [75], a state-of-the-art PnPDP method. STEP inherits the ability of DAPS to handle general inverse problems with nonlinear forward models and does not require heuristics to enforce temporal consistency.

We demonstrate the effectiveness of STEP on two challenging scientific video inverse problems: black hole video reconstruction (previewed in Figure 1) and dynamic MRI, where the underlying targets exhibit significantly different spatiotemporal characteristics. As Figure 1 illustrates, STEP not only achieves state-of-the-art results with improved spatial and temporal consistency but also effectively captures the multi-modal nature of highly ill-posed problems, recovering diverse plausible solutions from the posterior distribution. Notably, our approach achieves substantial improvements in both spatial and temporal consistency, significantly outperforming baselines across most evaluation metrics. For example, our method demonstrates a 1.57dB and 2.38dB improvement on PSNR on black hole video reconstruction and dynamic MRI, respectively. More importantly, for the challenging black hole video reconstruction, while baseline methods have difficulty accurately recovering temporal dynamics, our method demonstrates spatiotemporal structure that closely aligns with ground truth video (Fig. 4), under extremely sparse measurement conditions.

2 Background

Video inverse problems Existing approaches to video inverse problems have primarily focused on restoration and editing problems, such as super-resolution and JPEG artifact removal, for natural videos [9, 15, 35, 16, 82, 36]. Since it is computationally expensive to obtain a well-trained video diffusion model (VDM) for natural videos, existing methods instead rely on an image diffusion model (IDM) to process each video frame and propose various techniques to enforce temporal consistency. For example, the *batch-consistent sampling* (BCS) technique fixes the injected noise across all video frames [35, 36]. Another common approach, which we refer to as *noise warping*, first extracts optical flow from measurements and uses it to warp the injected noise across all video frames [9, 15, 73, 16]. As observed empirically, the dynamics of the injected noise translates into that of the generated video. These methods work well in restoration problems where the dynamics is mostly preserved

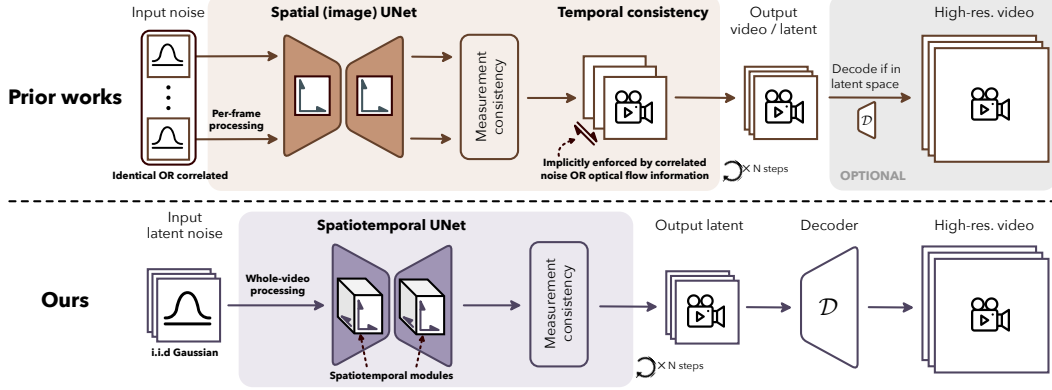


Figure 2: **A schematic comparison between prior works (top) and our STEP framework (bottom) for video inverse problems.** The bold texts highlight the key differences between them. While prior works use an image diffusion model and enforce temporal consistency using simple heuristics or warping noise with optical flow, we directly learn a spatiotemporal diffusion prior.

in the measurements or video editing tasks where the dynamics is given. However, these methods face challenges when dealing with scientific problems because the measurements may belong to a different domain that is nontrivial to invert, or substantial spatiotemporal information may be lost in the measurement process.

Plug-and-play diffusion priors for inverse problems Plug-and-play diffusion priors (PnPDP) constitute a family of methods that leverage diffusion models as priors for solving inverse problems. The goal is to draw samples from the posterior distribution $p(\mathbf{x}_0|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x}_0)p(\mathbf{x}_0)$ where the likelihood $p(\mathbf{y}|\mathbf{x}_0)$ can be derived from knowledge of the physical system, and the prior $p(\mathbf{x}_0)$ is characterized by a diffusion model. Various formulations have been considered in the existing literature, including measurement guidance [11, 43, 32, 46], variable splitting [14, 38, 45, 71, 70, 75], variational Bayes [40, 23, 22], and Sequential Monte Carlo (SMC) [68, 55, 8, 18]. A major advantage of PnPDP is its ability to handle inverse problems in a general way without task-specific design. Video inverse problems fit into the PnPDP framework in principle, but existing PnPDP methods have mainly focused on the static image setting. The main reason is that characterizing the prior distribution $p(\mathbf{x}_0)$ for videos is challenging due to their high dimensionality and a potentially limited number of samples for training. Prior work [34] has explored the applicability of PnPDP to an optical scattering problem with a pixel-space video diffusion prior. In this work, we develop a general framework that can efficiently handle VIPs with larger spatiotemporal dimensions.

Diffusion models as image and video priors Latent diffusion models (LDMs) [42] offer a way to reduce the computational cost otherwise necessary to model the original high-dimensional data distribution. LDMs generate an efficient, low-dimensional latent representation $\mathbf{z}_0$ of data $\mathbf{x}_0$ with a pretrained perceptual compression encoder $\mathcal{E}$ and decoder $\mathcal{D}$, which satisfy $\mathbf{z}_0 = \mathcal{E}(\mathbf{x}_0)$ and $\mathcal{D}(\mathbf{z}_0) \approx \mathbf{x}_0$. The compression models $\mathcal{E}$ and $\mathcal{D}$ can be trained as VAE variants [33, 41, 6] with KL divergence regularization or VQGAN variants [25, 20, 58] with quantization regularization. Video latent diffusion models are commonly believed to be hard to train due to the computational overhead of 3D modules and the requirement of a large video dataset [5, 80, 77, 27, 4]. Many recent methods tend to solve video modeling by fine-tuning from a pretrained IDM with a video dataset [67, 5, 4]. Our work leverages these techniques to solve scientific VIPs.

3 Method

In this section, we introduce our framework, STEP, for solving video inverse problems with **SpatioTemporal** diffusion **P**riors. Figure 2 illustrates the conceptual difference between our approach and the prior works on video inverse problems. Designed mainly for restoration problems on natural videos, prior works rely on IDMs to incorporate spatial prior and enforce temporal consistency by injecting correlated noise in the diffusion process. As we will show in the experiments, these techniques do not work well when dealing with ill-posed scientific problems. Our method

instead adopts a whole-video formulation (Sec. 3.1) and simultaneously handles spatial and temporal dimensions within a PnPDP framework (Sec. 3.2). The core of our method is to efficiently obtain spatiotemporal diffusion priors (Sec. 3.3), which directly learn temporal information from data and eliminate the need for designing task-specific temporal heuristics.

3.1 General approach

We consider the general formulation of video inverse problems where the goal is to recover a moving target $\mathbf{x}_0$ from the time-varying measurements

$$\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \mathbf{n}. \quad (1)$$

Here $\mathcal{A}(\cdot)$ is the forward model, which could be nonlinear, and $\mathbf{n}$ is the measurement noise. In many scientific problems, such as black hole video reconstruction, substantial spatiotemporal information of $\mathbf{x}_0$ is lost in the measurement process due to the ill-posed and potentially nonlinear nature of $\mathcal{A}(\cdot)$, necessitating a prior on both the spatial and temporal dimensions of $\mathbf{x}_0$ for meaningful recovery.

Instead of taking a per-frame processing approach using IDMs, we propose to directly learn the video distribution $p(\mathbf{x}_0)$ from data and draw samples from the posterior distribution $p(\mathbf{x}_0|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x}_0)p(\mathbf{x}_0)$ via the PnPDP framework. This enables us to deal with more general problems in which the extraction of temporal information directly from measurements is hard.

Solve video inverse problems in latent space. While the likelihood $p(\mathbf{y}|\mathbf{x}_0)$ can be derived from Equation (1), it is often challenging to characterize the prior distribution $p(\mathbf{x}_0)$ for videos due to its high dimensionality and potentially limited number of samples for training. To overcome these challenges, we propose to impose a spatiotemporal prior in the latent space. Assuming that the set of likely $\mathbf{x}_0$'s is in the range of a decoder $\mathcal{D}$, we have that $\exists \mathbf{z}_0$ s.t. $\mathbf{x}_0 = \mathcal{D}(\mathbf{z}_0)$ and can thus rewrite Equation (1) as:

$$\mathbf{y} = \mathcal{A}(\mathcal{D}(\mathbf{z}_0)) + \mathbf{n}. \quad (2)$$

It follows that the posterior $p(\mathbf{x}_0|\mathbf{y})$ is the pushforward of the latent posterior $p(\mathbf{z}_0|\mathbf{y})$ through $\mathcal{D}$, so it suffices to first generate latent samples from $p(\mathbf{z}_0|\mathbf{y})$ and then decode them by $\mathcal{D}$.

3.2 Decoupled Annealing Posterior Sampling

STEP leverages the PnPDP framework to sample the posterior distribution given by Equation (2). In this work, we employ the Decoupled Annealing Posterior Sampling (DAPS) method [75], which has demonstrated strong performance on various scientific inverse problems [79]. It is also easily compatible with latent diffusion models, making it a suitable choice for our purpose.

DAPS samples the target latent posterior $p(\mathbf{z}_0|\mathbf{y})$ by sequentially sampling $p(\mathbf{z}_t|\mathbf{y})$ from $t = T$ to $t = 0$. To do so, DAPS first generates a sample from $p(\mathbf{z}_T|\mathbf{y}) \approx \mathcal{N}(\mathbf{0}, \sigma_{\max}^2 \mathbf{I})$ and sequentially draws a sample from $p(\mathbf{z}_{t_{i-1}}|\mathbf{y})$ given a sample from $p(\mathbf{z}_{t_i}|\mathbf{y})$ for $i = N, \dots, 1$ based on a time schedule $\{t_i\}_{i=1}^N$. As shown by Proposition 1 of [75], this is possible if one can sample from

$$p(\mathbf{z}_0|\mathbf{z}_t, \mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{z}_0, \mathbf{z}_t)p(\mathbf{z}_0|\mathbf{z}_t)}{p(\mathbf{y}|\mathbf{z}_t)} \propto p(\mathbf{y}|\mathbf{z}_0)p(\mathbf{z}_0|\mathbf{z}_t).$$

Since we have access to the gradient of this unnormalized distribution, we can sample it using MCMC methods, such as Langevin Monte Carlo (LMC) [65] and Hamiltonian Monte Carlo (HMC) [2]. After obtaining $\hat{\mathbf{z}}_0 \sim p(\mathbf{z}_0|\mathbf{z}_t, \mathbf{y})$, we then sample from $p(\mathbf{z}_{t_{i-1}}|\mathbf{y})$ by sampling $\mathbf{z}_{t_{i-1}} \sim \mathcal{N}(\hat{\mathbf{z}}_0, \sigma_{t_{i-1}}^2 \mathbf{I})$ due to Proposition 1 of [75]. The pseudocode and more technical details of the STEP with DAPS are provided in Sec. A.

3.3 Efficient training of spatiotemporal diffusion priors

We propose to obtain spatiotemporal diffusion priors for scientific video inverse problems in an efficient way via the following three steps.

Step 1: Training a latent diffusion model (LDM) as an image prior We start by training a VAE [33] using the standard L_1 reconstruction loss with a scaled KL divergence loss on an image dataset. The KL divergence scaling factor is set to much less than 1 to prevent excessive regularization of the latent space. This allows us to obtain an image encoder $\mathcal{E}$ and decoder $\mathcal{D}$. Once they are trained, we fix their parameters and train a 2D UNet model $s_\theta(\mathbf{z}_t; \sigma_t)$ using the standard denoising score matching loss [59]. Despite recent progress in 3D spatiotemporal encoders and decoders [72, 69, 10], we opt for a 2D spatial encoder and decoder to process each frame independently. This choice is due to efficiency considerations, as the decoder $\mathcal{D}$ will be called repeatedly in DAPS during inference.

Step 2: Turning an image prior into a spatiotemporal prior Upon obtaining an LDM as an image prior, we use a spatiotemporal UNet architecture to parameterize the time-dependent video score function, i.e. $s_\theta(\mathbf{z}_t; \sigma_t) \approx \nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t; \sigma_t)$, leveraging recent advancements in video generation [62, 27]. The key component in the architecture is the spatiotemporal module for 3D modeling, as illustrated in Figure 3. Given a pretrained IDM with a 2D UNet, we introduce a zero-initialized temporal module for each 2D spatial module within the UNet. Specifically, for an input feature f_{in} , let f_{out} be the output of the spatiotemporal module, with f_{spat} and f_{temp} representing the outputs of the spatial and temporal branches, respectively. These features are combined using an α -blending mechanism:

$$f_{\text{out}} = (1 - \alpha) \cdot f_{\text{spat}} + \alpha \cdot f_{\text{temp}}, \quad (3)$$

where $\alpha \in \mathbb{R}$ is a learnable parameter initialized as 0 in each spatiotemporal module. This design allows us to inherit the weights of the 2D spatial modules from the pretrained IDM, significantly reducing training time. Additionally, by factorizing the 3D module into a 2D spatial module and a 1D temporal module, the spatiotemporal UNet has marginal computational overhead compared to the original 2D UNet, striking a good balance between model capacity and efficiency.

Step 3: Image-video joint fine-tuning To further improve performance and be compatible with both image and video inputs, we introduce an ON/OFF switch signal in the spatiotemporal module. When the switch is set to OFF (indicating image inputs), the temporal module is disabled (or equivalently $\alpha = 0$). This ensures that $f_{\text{out}} = f_{\text{spat}}$ and reduces the spatiotemporal module to the original 2D spatial module, which processes each frame independently. During training, we initialize the weights of the spatial modules based on a pretrained image diffusion model and fine-tune all parameters of the spatiotemporal UNet using both image and video data. During fine-tuning, the model receives video data with probability $p_{\text{joint}} \in [0, 1]$ and receives images (with switch set to OFF accordingly) with probability $1 - p_{\text{joint}}$. The probability p_{joint} is a tunable hyperparameter controlling the proportion of real video data in training. Pseudo video regularization helps the spatiotemporal UNet retain the spatial capabilities of the initialized spatial UNet. This strategy, proven effective in previous work [62], stabilizes training and prevents overfitting to the video dataset.

4 Experiments

4.1 Tasks and setup

We consider two scientific video inverse problems: black hole video reconstruction [21] and dynamic MRI [24]. Although both are scientific imaging tasks, they have significantly different characteristics. Black hole video reconstruction involves simple spatial structures (usually a ring structure) but

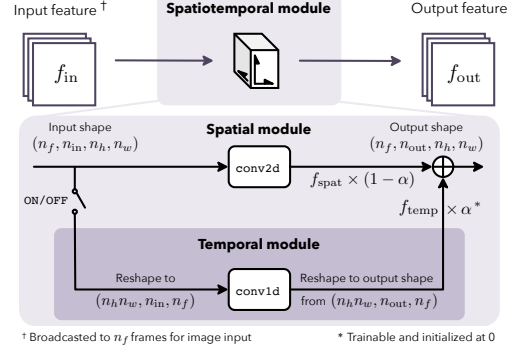


Figure 3: **Architecture of the spatiotemporal module.** Given a pretrained image diffusion UNet, we add a zero-initialized temporal module with an ON/OFF switch to each 2D spatial module and initialize the additive weight α to zero. Thus, it will add no effect at the start of fine-tuning and gradually learn by the video training data. The number of frames, height, and width are denoted by n_f , n_h , and n_w , respectively. The numbers of channels for input features (f_{in}) and output features (f_{out}) are denoted by n_{in} and n_{out} , respectively.

complex temporal dynamics that obey physical constraints. In contrast, dynamic MRI requires higher spatial fidelity with relatively simpler temporal dynamics, such as periodic heartbeat motion.

Black hole video reconstruction We consider the problem of observing the Sagittarius A* black hole using the Event Horizon Telescope (EHT) array in 2017 [21]. The measurements consist of sparse vectors with a dimensionality of 1,856 derived from an underlying black hole video comprising 64 frames, each with a spatial resolution of 256×256 pixels, capturing the black hole’s dynamics during the 100-minute observation period. These measurements are given by calculating the closure quantities based on the complex visibilities (a detailed description of the problem is available in Appendix B.1). For the training dataset, we employ the *general relativistic magnetohydrodynamic (GRMHD)* simulations [66] of Sagittarius A*, covering various black hole models and observational conditions. The dataset comprises 648 simulated black hole videos and 50,000 black hole images.

Dynamic MRI We consider a standard compressed sensing MRI setup with two acceleration scenarios: $8 \times$ acceleration with 12 auto-calibration signal (ACS) lines and $6 \times$ acceleration with 24 ACS lines. Further details are provided in Appendix B.2. We utilize the publicly available cardiac cine MRI dataset from the CMRxRecon Challenge 2023 [61] for training. This dataset includes 3,324 cardiac MRI sequences, each containing fully sampled, ECG-triggered k -space data from 300 patients, featuring various canonical cardiac imaging views. Each sequence is processed into a video consisting of 12 frames with a spatial resolution of 192×192 pixels. We construct the image dataset by extracting the individual frames from the videos, resulting in 39,888 images in total.

4.2 Baselines and our method

Recall from Sec. 3 that our method leverages a learned spatiotemporal prior to avoid explicitly estimating temporal dynamics from measurements at inference time. To evaluate the effectiveness of this strategy, we compare our method to existing IDM-based approaches. Based on how temporal information is incorporated, we categorize these baseline methods into two groups as follows.

Group 1: simple heuristics We consider two baselines: BIS [36] and BCS [36]. *Batch independent sampling* (BIS) reconstructs each frame independently, while *batch consistent sampling* (BCS) implicitly incorporates a static temporal prior to enforce consistency across frames.

Group 2: noise warping We include another line of baselines: $\mathcal{I}$ -noise [9, 16], and GP-Warp [15]. These baselines enforce temporal consistency by warping the noise using optical flow estimated directly from the measurements. Besides, we include more conventional warping strategies such as *Bilinear*, *Bicubic*, *Nearest* from work [9]. For the black hole video reconstruction task, a direct inversion from the sparse measurement vector is infeasible. Thus, to demonstrate the upper-bound performance of these methods, we leverage the ground truth video as an oracle to derive the optical flow. For the dynamic MRI task, we utilize a naive inversion obtained via inverse Fourier transformation to estimate the optical flow. We follow the methodology of [9], leveraging a pretrained model [54] to extract optical flow.

Ours: two variants of STEP We evaluate two variants differentiated by their spatiotemporal priors: STEP (video-only), which uses a spatiotemporal prior trained exclusively on video data, and STEP (image-video joint), which initializes with a pre-trained IDM and undergoes joint image-video fine-tuning. We follow the Sec. C to train our priors until convergence. The detailed training hyperparameters are summarized in Table 3. To make a fair comparison, we run all the baselines with DAPS [75]. The detailed implementation of each baseline can be found at Appendix A and B.3.

4.3 Metrics

We evaluate our results on three aspects: (1) spatial similarity, (2) temporal consistency, and (3) measurement data fit.

Spatial Spatial similarity is assessed by calculating Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [64], and Learned Perceptual Image Patch Similarity (LPIPS) [76]. Metrics are computed per frame and averaged across all frames. We utilize implementations from

Table 1: **Quantitative results on black hole video reconstruction and dynamic MRI.** We compare our method against baselines by reporting the mean and standard deviation (shown in parentheses) of selected evaluation metrics computed over 10 test videos (FVD is reported without standard deviation since it evaluates the set of 10 videos collectively). The results clearly demonstrate that by leveraging the spatiotemporal prior, STEP consistently achieves improvements in both spatial quality and temporal consistency relative to baseline methods.

Tasks	Methods	PSNR ($\uparrow$)	SSIM ($\uparrow$)	LPIPS ($\downarrow$)	d-PSNR ($\uparrow$)	d-SSIM ($\uparrow$)	FVD ($\downarrow$)	Data Misfit ($\downarrow$)
Black hole	BIS [36]	23.79 (1.41)	0.718 (0.047)	0.179 (0.031)	29.26 (1.51)	0.938 (0.015)	1429.42	1.719 (1.277)
	BCS [36]	27.66 (2.04)	<u>0.816</u> (0.053)	0.124 (0.040)	<u>41.71</u> (1.99)	0.979 (0.008)	564.43	1.426 (0.784)
	Bilinear [9]	26.11 (2.05)	0.718 (0.067)	0.151 (0.044)	33.14 (2.33)	0.958 (0.013)	1335.95	1.384 (0.742)
	Bicubic [9]	25.68 (1.88)	0.730 (0.058)	0.163 (0.041)	31.70 (1.88)	0.952 (0.015)	1521.14	1.736 (1.259)
	Nearest [9]	25.29 (1.80)	0.754 (0.059)	0.164 (0.042)	30.76 (1.75)	0.943 (0.017)	1171.07	1.691 (1.020)
	ζ -noise [9, 16]	24.90 (1.52)	0.745 (0.058)	0.163 (0.034)	31.87 (1.93)	0.945 (0.017)	1253.81	1.655 (1.020)
	GP-Warp [15]	23.98 (1.28)	0.721 (0.043)	0.176 (0.029)	29.21 (1.24)	0.938 (0.014)	1395.15	1.721 (1.225)
	STEP (video only)	28.71 (1.81)	0.802 (0.079)	0.120 (0.041)	41.41 (2.39)	0.975 (0.011)	238.36	<u>1.124</u> (0.136)
	STEP (image-video joint)	30.28 (2.71)	0.865 (0.063)	0.095 (0.039)	42.09 (2.63)	<u>0.976</u> (0.011)	170.67	1.114 (0.154)
MRI (8 $\times$)	BIS [36]	35.04 (0.76)	0.889 (0.016)	0.100 (0.012)	38.91 (1.13)	0.918 (0.012)	82.30	9.203 (0.698)
	BCS [36]	35.43 (0.97)	0.893 (0.016)	0.099 (0.012)	39.91 (1.09)	0.931 (0.011)	95.68	9.225 (0.693)
	Bilinear [9]	35.30 (0.96)	0.896 (0.016)	0.098 (0.012)	39.87 (1.14)	0.931 (0.011)	87.90	9.157 (0.670)
	Bicubic [9]	35.31 (0.91)	0.896 (0.016)	0.098 (0.011)	40.11 (1.10)	0.934 (0.010)	108.90	9.189 (0.653)
	Nearest [9]	34.87 (0.90)	0.895 (0.018)	0.099 (0.013)	40.09 (1.14)	0.933 (0.011)	108.57	9.188 (0.656)
	ζ -noise [9, 16]	35.55 (1.03)	0.892 (0.020)	0.099 (0.014)	39.77 (1.36)	0.929 (0.015)	89.59	9.208 (0.693)
	GP-Warp [15]	34.49 (0.65)	0.886 (0.016)	0.102 (0.012)	38.71 (1.14)	0.916 (0.013)	92.19	9.209 (0.659)
	STEP (video only)	37.00 (1.46)	0.927 (0.019)	0.086 (0.013)	43.50 (2.70)	0.963 (0.015)	75.27	8.817 (0.650)
	STEP (image-video joint)	39.38 (1.16)	0.951 (0.009)	0.078 (0.011)	44.86 (1.92)	0.974 (0.006)	<u>78.60</u>	8.753 (0.603)

piq [31], normalizing images to the range $[0, 1]$. For grayscale images, frames are replicated across three channels before computing LPIPS scores.

Temporal Temporal consistency is quantified using delta-based PSNR (d-PSNR) and delta-based SSIM (d-SSIM), measuring the similarity of normalized differences between consecutive frames, with results averaged over all frames. Additionally, we compute the Fréchet Video Distance (FVD) [56] between the reconstructed videos and our test dataset to evaluate distributional similarity.

Data fit Lastly, we report measurement data fit using task-specific metrics. For the black hole video reconstruction task, we employ the unified average $\tilde{\chi}^2$ statistic (defined in Equation (13) in Appendix), where values closer to 1 indicate better data fidelity. For dynamic MRI, we measure data misfit by computing the mean squared error in measurement space.

4.4 Main results ¹

STEP reconstructs videos with better spatial and temporal coherence. We provide quantitative results in Tab. 1 and visual comparisons in Fig. 4 and Fig. 5. STEP outperforms baselines in overall video quality across most metrics and generates video reconstructions with significantly better spatiotemporal coherence. Fig. 4 visualizes an x - t slice, representing the temporal evolution (horizontal axis) of a spatial slice (marked by the cyan vertical line). We find that the noise warping baselines fail to constrain temporal consistency, as indicated by the fluctuating ring diameters in the x - t slices, while BCS provides almost static reconstructions. In contrast, STEP exhibits closer alignment with the ground truth and improved temporal consistency. This is further illustrated by the averaged optical flow visualization, highlighting that STEP faithfully captures the underlying temporal dynamic. Similar trends can be observed in Fig. 5.

Noise warping is less effective in scientific problems. We observe that the performance of noise-warping strategies in challenging scientific VIPs does not correlate with noise-warping accuracy. For example, although the ζ -noise approach has demonstrated superior noise-warping capability given optical flow (as shown in [9]), it underperforms simpler strategies such as *Bilinear* and *Bicubic* interpolation in challenging VIPs. This discrepancy arises primarily due to: (1) inaccuracies in the optical flow, and (2) the inherent difficulty of effectively manipulating noise in latent space through pixel space optical-flow-guided warping. Consequently, methods relying on precise optical flow and carefully designed warping strategies struggle with tasks requiring high temporal consistency, such as

¹For better assessment of video reconstruction quality, we strongly encourage the readers to watch our supplementary video.

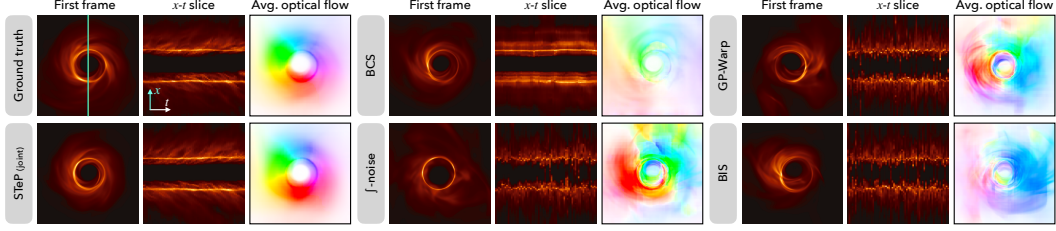


Figure 4: **Visual examples of STEP (bottom left) and baselines for black hole video reconstruction.** To facilitate analysis of the reconstructed spatiotemporal structures, we present results in three ways: (1) a single frame to illustrate spatial fidelity, (2) an x - t slice depicting temporal evolution of a vertical line to evaluate temporal consistency, and (3) the averaged optical flow visualized using the standard color scheme from [54] to assess spatiotemporal coherence jointly. Compared to baselines, STEP exhibits clearer alignment with ground truth videos across all aspects.

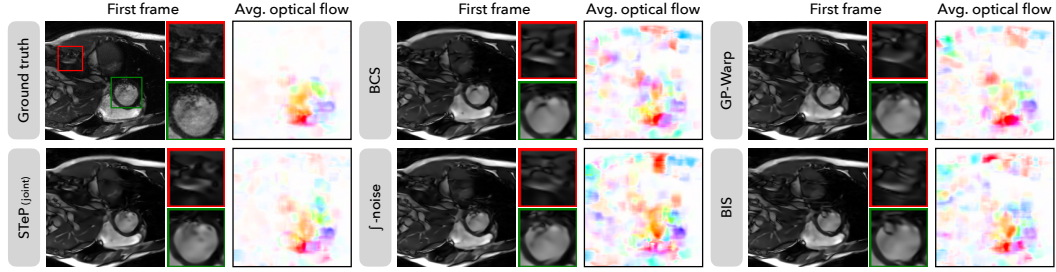


Figure 5: **Visual examples of STEP (bottom left) and baselines for dynamic MRI.** We visualize a representative frame along with two zoomed-in regions for each method to better illustrate spatial fidelity. Benefiting from its robust spatiotemporal prior, STEP provide reconstructions with fewer structural artifacts and temporal fluctuations, as indicated by its averaged optical flow aligning more closely with the ground truth. This demonstrates the effectiveness of our learned spatiotemporal prior in enhancing both spatial and temporal consistency.

black hole video reconstruction. In contrast, our approach employs a data-driven spatiotemporal prior learned from training data, effectively overcoming these limitations and enabling broader applicability to scientific tasks with significant temporal uncertainty.

Joint fine-tuning benefits both spatial and temporal consistency. As shown in Tab. 1, STEP with an image-video jointly fine-tuned spatiotemporal prior outperforms both IDM-based baselines and STEP trained video data only. To further illustrate these improvements, we provide a detailed visual comparison in Fig. 6. In Fig. 6 (a), we analyze the temporal dynamics using averaged delta frames, which visualize the averaged differences between consecutive frames over a window. We observe that delta frames from BCS remain largely unchanged as the temporal window expands, indicating limited temporal variation. In contrast, STEP (joint) effectively captures more complex and coherent temporal dynamics. In Fig. 6 (b), we compare STEP (joint) and STEP (video only) with a focus on spatial fidelity. Although their reconstructions have similar average optical flow, STEP (joint) have fewer artifacts and hallucinations as pointed out by the arrows. Such an improvement highlights the value of using image data in both the pre-training and fine-tuning stages.

STEP provides diverse and equally plausible solutions. Due to the highly ill-posed and extremely sparse measurements in black hole video reconstruction, STEP can produce multiple semantically diverse reconstructions, each visually plausible and consistent with measurement data. Since the true posterior distribution is unknown, directly quantifying mode coverage is infeasible. Instead, we draw 100 *i.i.d.* samples and cluster them based on spatial appearance and temporal dynamics. As illustrated in Fig. 1, we identify three distinct modes with nearly identical data fidelity (see Tab. 5 exact data misfit values). One recovered mode aligns closely with the ground truth, while the others differ in rotation direction or spatial structure. This demonstrates that our method not only accurately reconstructs the underlying ground truth but also discovers additional plausible solutions.

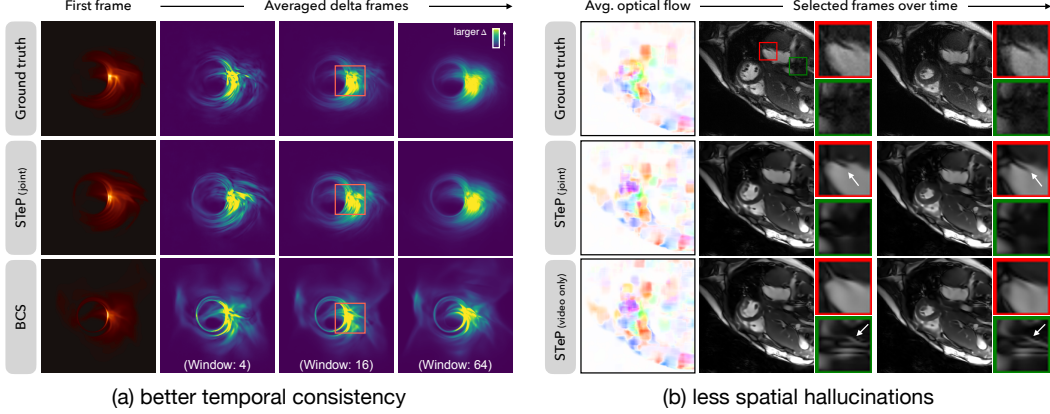


Figure 6: **Detailed comparison on black hole video reconstruction and dynamic MRI.** Left: We compare STeP (joint) and the BCS baseline [36] by visualizing the averaged delta frames (difference images) over an expanding window. The delta frames given by STeP (joint) better align with the ground truth, indicating better temporal consistency. Right: We also compare the spatial fidelity between STeP (joint) and its variant STeP (video-only). Trained on both images and videos, STeP (joint) provide reconstructions with less spatial hallucinations compared to STeP (video-only).

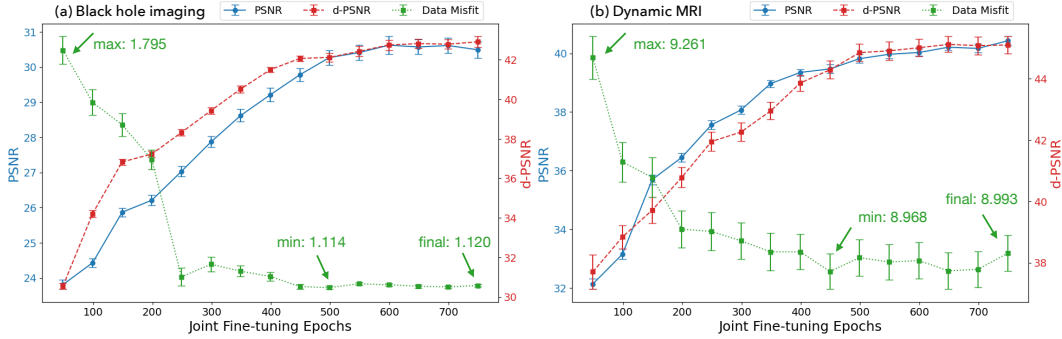


Figure 7: **Consistent improvement in image-video joint fine-tuning.** We evaluate intermediate checkpoints of (a) black hole video reconstruction and (b) dynamic MRI ($8\times$). Spatial similarity (measured by PSNR), temporal consistency (measured by d-PSNR) and measurement data fit (measured by data misfit) show steady improvement.

4.5 Ablation on image-video joint finetuning

To show the effectiveness of the proposed image-video joint finetuning technique, we show the quantitative results of using checkpoints of the spatiotemporal UNet that were fine-tuned for different numbers of epochs. We assess performance using PSNR (blue curve), d-PSNR (red curve), and a data-fitting metric (green curve), as shown in Fig. 7. Since the spatiotemporal UNet is initialized from a pretrained image diffusion model, these curves indicate steady improvement in spatiotemporal consistency and data fitting as the prior is fine-tuned.

5 Conclusion

We introduced STeP, a framework designed for scientific video inverse problems (VIPs) that integrates a spatiotemporal diffusion prior into a PnPDP solver. By explicitly modeling spatiotemporal structures, our method eliminates the reliance on temporal heuristics and recovers coherent videos from sparse measurements. We evaluated STeP on two challenging scientific VIPs: black hole video reconstruction and dynamic MRI. STeP consistently outperformed existing methods, achieving better spatial fidelity and more accurate temporal consistency. Our framework demonstrates a general and scalable approach to scientific VIPs by combining sophisticated spatiotemporal diffusion priors with state-of-the-art PnPDP methods.

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References

- [1] Kazunori Akiyama, Antxon Alberdi, Walter Alef, Keiichi Asada, Rebecca Azulay, Anne-Kathrin Bacsko, David Ball, Mislav Baloković, John Barrett, Dan Bintley, et al. First m87 event horizon telescope results. iv. imaging the central supermassive black hole. *The Astrophysical Journal Letters*, 875(1):L4, 2019.
- [2] Michael Betancourt and Mark Girolami. Hamiltonian monte carlo for hierarchical models. *Current trends in Bayesian methodology with applications*, 79(30):2–4, 2015.
- [3] Lindy Blackburn, Dominic W. Pesce, Michael D. Johnson, Maciek Wielgus, Andrew A. Chael, Pierre Christian, and Sheperd S. Doleman. Closure statistics in interferometric data. *The Astrophysical Journal*, 894(1):31, may 2020.
- [4] Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, Varun Jampani, and Robin Rombach. Stable video diffusion: Scaling latent video diffusion models to large datasets, 2023.
- [5] Andreas Blattmann, Robin Rombach, Huan Ling, Tim Dockhorn, Seung Wook Kim, Sanja Fidler, and Karsten Kreis. Align your latents: High-resolution video synthesis with latent diffusion models, 2023.
- [6] Christopher P. Burgess, Irina Higgins, Arka Pal, Loic Matthey, Nick Watters, Guillaume Desjardins, and Alexander Lerchner. Understanding disentangling in β -vae, 2018.
- [7] John Charles Butcher. Numerical methods for ordinary differential equationsnumerical. 2008.
- [8] Gabriel Victorino Cardoso, Yazid Janati, Sylvain Le Corff, and Éric Moulines. Monte carlo guided diffusion for bayesian linear inverse problems. *ArXiv*, abs/2308.07983, 2023.
- [9] Pascal Chang, Jingwei Tang, Markus Gross, and Vinicius C. Azevedo. How i warped your noise: a temporally-correlated noise prior for diffusion models. In *The Twelfth International Conference on Learning Representations*, 2024.
- [10] Liuhan Chen, Zongjian Li, Bin Lin, Bin Zhu, Qian Wang, Shenghai Yuan, Xing Zhou, Xinhua Cheng, and Li Yuan. Od-vae: An omni-dimensional video compressor for improving latent video diffusion model, 2024.
- [11] Hyungjin Chung, Jeongsol Kim, Michael Thompson Mccann, Marc Louis Klasky, and Jong Chul Ye. Diffusion posterior sampling for general noisy inverse problems. In *The Eleventh International Conference on Learning Representations*, 2023.
- [12] Hyungjin Chung, Eun Sun Lee, and Jong Chul Ye. Mr image denoising and super-resolution using regularized reverse diffusion. *IEEE Transactions on Medical Imaging*, 42(4):922–934, 2022.
- [13] Hyungjin Chung and Jong Chul Ye. Score-based diffusion models for accelerated mri. *Medical Image Analysis*, 80:102479, 05 2022.
- [14] Florentin Coeurdoux, Nicolas Dobigeon, and Pierre Chainais. Plug-and-play split gibbs sampler: embedding deep generative priors in bayesian inference, 2023.
- [15] Giannis Daras, Weili Nie, Karsten Kreis, Alex Dimakis, Morteza Mardani, Nikola Borislavov Kovachki, and Arash Vahdat. Warped diffusion: Solving video inverse problems with image diffusion models. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- [16] Yitong Deng, Winnie Lin, Lingxiao Li, Dmitriy Smirnov, Ryan D Burgert, Ning Yu, Vincent Dedun, and Mohammad H. Taghavi. Infinite-resolution integral noise warping for diffusion models. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [17] Zolnamar Dorjsembe, Hsing-Kuo Pao, Sotavilan Odonchimed, and Furen Xiao. Conditional diffusion models for semantic 3d brain mri synthesis. *IEEE Journal of Biomedical and Health Informatics*, 2024.

- [18] Zehao Dou and Yang Song. Diffusion posterior sampling for linear inverse problem solving: A filtering perspective. In *The Twelfth International Conference on Learning Representations*, 2024.
- [19] The Event Horizon Telescope Collaboration EHTC. First m87 event horizon telescope results. iii. data processing and calibration. *The Astrophysical Journal Letters*, 875(1):L3, apr 2019.
- [20] Patrick Esser, Robin Rombach, and Björn Ommer. Taming transformers for high-resolution image synthesis, 2021.
- [21] Event Horizon Telescope Collaboration. First Sagittarius A* Event Horizon Telescope Results. III. Imaging of the Galactic Center Supermassive Black Hole. *The Astrophysical Journal Letters*, 930(2):L14, May 2022.
- [22] Berthy Feng and Katherine Bouman. Variational bayesian imaging with an efficient surrogate score-based prior. *Transactions on Machine Learning Research*, 2024.
- [23] Berthy T. Feng, Jamie Smith, Michael Rubinstein, Huiwen Chang, Katherine L. Bouman, and William T. Freeman. Score-based diffusion models as principled priors for inverse imaging. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10520–10531, October 2023.
- [24] Urs Gamper, Peter Boesiger, and Sebastian Kozerke. Compressed sensing in dynamic mri. *Magnetic Resonance in Medicine*, 59(2):365–373, 2008.
- [25] Shuyang Gu, Dong Chen, Jianmin Bao, Fang Wen, Bo Zhang, Dongdong Chen, Lu Yuan, and Baining Guo. Vector quantized diffusion model for text-to-image synthesis, 2022.
- [26] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- [27] Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. Video diffusion models, 2022.
- [28] Alex Ling Yu Hung, Kai Zhao, Haoxin Zheng, Ran Yan, Steven S Raman, Demetri Terzopoulos, and Kyunghyun Sung. Med-diff: Conditional medical image generation with diffusion models. *Bioengineering*, 10(11):1258, 2023.
- [29] Ajil Jalal, Marius Arvinte, Giannis Daras, Eric Price, Alex Dimakis, and Jonathan Tamir. Robust compressed sensing MRI with deep generative priors. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021.
- [30] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, 2022.
- [31] Sergey Kastryulin, Jamil Zakirov, Denis Prokopenko, and Dmitry V. Dylov. Pytorch image quality: Metrics for image quality assessment, 2022.
- [32] Bahjat Kavar, Michael Elad, Stefano Ermon, and Jiaming Song. Denoising diffusion restoration models. *Advances in Neural Information Processing Systems*, 35:23593–23606, 2022.
- [33] Diederik P Kingma and Max Welling. Auto-encoding variational bayes, 2022.
- [34] Taesung Kwon, Gookho Song, Yoosun Kim, Jong Chul Ye, and Mooseok Jang. Seeing video through optical scattering media using spatio-temporal diffusion models, 2024.
- [35] Taesung Kwon and Jong Chul Ye. Vision-xl: High definition video inverse problem solver using latent image diffusion models, 2024.
- [36] Taesung Kwon and Jong Chul Ye. Solving video inverse problems using image diffusion models. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [37] Aviad Levis, Daeyoung Lee, Joel A. Tropp, Charles F. Gammie, and Katherine L. Bouman. Inference of black hole fluid-dynamics from sparse interferometric measurements. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 2320–2329, 2021.
- [38] Xiang Li, Soo Min Kwon, Ismail R. Alkhouri, Saiprasad Ravishankar, and Qing Qu. Decoupled data consistency with diffusion purification for image restoration, 2024.
- [39] Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. Repaint: Inpainting using denoising diffusion probabilistic models, 2022.

- [40] Morteza Mardani, Jiaming Song, Jan Kautz, and Arash Vahdat. A variational perspective on solving inverse problems with diffusion models. *arXiv preprint arXiv:2305.04391*, 2023.
- [41] Danilo Jimenez Rezende, Shakir Mohamed, and Daan Wierstra. Stochastic backpropagation and approximate inference in deep generative models, 2014.
- [42] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.
- [43] Litu Rout, Negin Raoof, Giannis Daras, Constantine Caramanis, Alex Dimakis, and Sanjay Shakkottai. Solving linear inverse problems provably via posterior sampling with latent diffusion models. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.
- [44] Chitwan Saharia, Jonathan Ho, William Chan, Tim Salimans, David J. Fleet, and Mohammad Norouzi. Image super-resolution via iterative refinement. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4):4713–4726, 2023.
- [45] Bowen Song, Soo Min Kwon, Zecheng Zhang, Xinyu Hu, Qing Qu, and Liyue Shen. Solving inverse problems with latent diffusion models via hard data consistency, 2024.
- [46] Jiaming Song, Arash Vahdat, Morteza Mardani, and Jan Kautz. Pseudoinverse-guided diffusion models for inverse problems. In *International Conference on Learning Representations*, 2023.
- [47] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *Neural Information Processing Systems*, 2019.
- [48] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution, 2020.
- [49] Yang Song, Liyue Shen, Lei Xing, and Stefano Ermon. Solving inverse problems in medical imaging with score-based generative models. In *International Conference on Learning Representations*, 2022.
- [50] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.
- [51] He Sun and Katherine L. Bouman. Deep probabilistic imaging: Uncertainty quantification and multi-modal solution characterization for computational imaging, 2020.
- [52] Yu Sun, Zihui Wu, Yifan Chen, Berthy Feng, and Katherine L. Bouman. Provable probabilistic imaging using score-based generative priors. *ArXiv*, abs/2310.10835, 2023.
- [53] Yu Sun, Zihui Wu, Yifan Chen, Berthy T. Feng, and Katherine L. Bouman. Provable probabilistic imaging using score-based generative priors. *IEEE Transactions on Computational Imaging*, 10:1290–1305, 2024.
- [54] Zachary Teed and Jia Deng. Raft: Recurrent all-pairs field transforms for optical flow, 2020.
- [55] Brian L. Trippe, Jason Yim, Doug Tischer, David Baker, Tamara Broderick, Regina Barzilay, and Tommi S. Jaakkola. Diffusion probabilistic modeling of protein backbones in 3d for the motif-scaffolding problem. In *The Eleventh International Conference on Learning Representations*, 2023.
- [56] Thomas Unterthiner, Sjoerd Van Steenkiste, Karol Kurach, Raphael Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric & challenges. *arXiv preprint arXiv:1812.01717*, 2018.
- [57] Pieter Hendrik van Cittert. Die wahrscheinliche schwingungsverteilung in einer von einer lichtquelle direkt oder mittels einer linse beleuchteten ebene. *Physica*, 1(1-6):201–210, 1934.
- [58] Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning, 2018.
- [59] Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23:1661–1674, 2011.
- [60] Patrick von Platen, Suraj Patil, Anton Lozhkov, Pedro Cuenca, Nathan Lambert, Kashif Rasul, Mishig Davaadorj, Dhruv Nair, Sayak Paul, William Berman, Yiyi Xu, Steven Liu, and Thomas Wolf. Diffusers: State-of-the-art diffusion models. <https://github.com/huggingface/diffusers>, 2022.

- [61] Chengyan Wang, Jun Lyu, Shuo Wang, Chen Qin, Kunyuan Guo, Xinyu Zhang, Xiaotong Yu, Yan Li, Fanwen Wang, Jianhua Jin, Zhang Shi, Ziqiang Xu, Yapeng Tian, Sha Hua, Zhensen Chen, Meng Liu, Mengting Sun, Xutong Kuang, Kang Wang, and Xiaobo Qu. Cmrrecon: A publicly available k-space dataset and benchmark to advance deep learning for cardiac mri. *Scientific Data*, 11, 06 2024.
- [62] Jiuniu Wang, Hangjie Yuan, Dayou Chen, Yingya Zhang, Xiang Wang, and Shiwei Zhang. Modelscape text-to-video technical report, 2023.
- [63] Yinhuai Wang, Jiwen Yu, and Jian Zhang. Zero-shot image restoration using denoising diffusion null-space model. *The Eleventh International Conference on Learning Representations*, 2023.
- [64] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.
- [65] Max Welling and Yee W Teh. Bayesian learning via stochastic gradient langevin dynamics. In *Proceedings of the 28th international conference on machine learning (ICML-11)*, pages 681–688. Citeseer, 2011.
- [66] George N Wong, Ben S Prather, Vedant Dhruv, Benjamin R Ryan, Monika Mościbrodzka, Chi-kwan Chan, Abhishek V Joshi, Ricardo Yarza, Angelo Ricarte, Hotaka Shiokawa, et al. Patoka: Simulating electromagnetic observables of black hole accretion. *The Astrophysical Journal Supplement Series*, 259(2):64, 2022.
- [67] Jay Zhangjie Wu, Yixiao Ge, Xintao Wang, Weixian Lei, Yuchao Gu, Yufei Shi, Wynne Hsu, Ying Shan, Xiaohu Qie, and Mike Zheng Shou. Tune-a-video: One-shot tuning of image diffusion models for text-to-video generation, 2023.
- [68] Luhuan Wu, Brian Trippe, Christian Naesseth, David Blei, and John P Cunningham. Practical and asymptotically exact conditional sampling in diffusion models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 31372–31403. Curran Associates, Inc., 2023.
- [69] Pingyu Wu, Kai Zhu, Yu Liu, Liming Zhao, Wei Zhai, Yang Cao, and Zheng-Jun Zha. Improved video vae for latent video diffusion model, 2024.
- [70] Zihui Wu, Yu Sun, Yifan Chen, Bingliang Zhang, Yisong Yue, and Katherine L Bouman. Principled probabilistic imaging using diffusion models as plug-and-play priors. *arXiv preprint arXiv:2405.18782*, 2024.
- [71] Xingyu Xu and Yuejie Chi. Provably robust score-based diffusion posterior sampling for plug-and-play image reconstruction, 2024.
- [72] Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang, Wenyi Hong, Xiaohan Zhang, Guanyu Feng, Da Yin, Xiaotao Gu, Yuxuan Zhang, Weihang Wang, Yean Cheng, Ting Liu, Bin Xu, Yuxiao Dong, and Jie Tang. Cogvideox: Text-to-video diffusion models with an expert transformer, 2024.
- [73] Chang-Han Yeh, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, Ting-Hsuan Chen, Hau-Shiang Shiu, and Yu-Lun Liu. Diffir2vr-zero: Zero-shot video restoration with diffusion-based image restoration models, 2024.
- [74] Frederik Zernike. The concept of degree of coherence and its application to optical problems. *Physica*, 5(8):785–795, 1938.
- [75] Bingliang Zhang, Wenda Chu, Julius Berner, Chenlin Meng, Anima Anandkumar, and Yang Song. Improving diffusion inverse problem solving with decoupled noise annealing. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [76] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595, Los Alamitos, CA, USA, jun 2018. IEEE Computer Society.
- [77] Zhixing Zhang, Bichen Wu, Xiaoyan Wang, Yaqiao Luo, Luxin Zhang, Yinan Zhao, Peter Vajda, Dimitris Metaxas, and Licheng Yu. Avid: Any-length video inpainting with diffusion model, 2024.
- [78] Hongkai Zheng, Wenda Chu, Austin Wang, Nikola Kovachki, Ricardo Baptista, and Yisong Yue. Ensemble kalman diffusion guidance: A derivative-free method for inverse problems, 2024.

- [79] Hongkai Zheng, Wenda Chu, Bingliang Zhang, Zihui Wu, Austin Wang, Berthy Feng, Caifeng Zou, Yu Sun, Nikola Borislavov Kovachki, Zachary E Ross, Katherine Bouman, and Yisong Yue. Inversebench: Benchmarking plug-and-play diffusion priors for inverse problems in physical sciences. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [80] Daquan Zhou, Weimin Wang, Hanshu Yan, Weiwei Lv, Yizhe Zhu, and Jiashi Feng. Magicvideo: Efficient video generation with latent diffusion models, 2023.
- [81] Yuanzhi Zhu, Kai Zhang, Jingyun Liang, Jiezhong Cao, Bihan Wen, Radu Timofte, and Luc Van Gool. Denoising diffusion models for plug-and-play image restoration. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops (NTIRE)*, 2023.
- [82] Zihao Zou, Jiaming Liu, Shirin Shoushtari, Yubo Wang, and Ulugbek S. Kamilov. Flair: A conditional diffusion framework with applications to face video restoration. In *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, pages 5228–5238, February 2025.

A Detailed implementation of STEP

Here, we summarize the proposed framework for solving video inverse problems in Algorithm 1.

Algorithm 1 STEP: a framework for solving video inverse problems with SpatioTemporal Prior

Require: Discretization time steps $\{t_i\}_{i=1}^N$ where $t_0 = 0$ and $t_N = T$, noise schedule σ_t , likelihood $p(\mathbf{y} \mid \cdot)$ with measurement $\mathbf{y}$, HMC step size η and damping factor γ , number of HMC updates M , pretrained latent score function $\mathbf{s}_\theta(\mathbf{z}; \sigma) \approx \nabla_{\mathbf{z}} \log p(\mathbf{z}; \sigma)$ with image decoder $\mathcal{D}$.

- 1: $\mathbf{z}_{t_N} \sim \mathcal{N}(\mathbf{0}, \sigma_{t_N}^2 \mathbf{I})$ ▷ Initialization
- 2: **for** $i = N, \dots, 1$ **do**
- 3: $\hat{\mathbf{z}}_0 \leftarrow \text{Backward}(\mathbf{z}_{t_i}; \mathbf{s}_\theta)$ ▷ Solve PF-ODE (4) backward from $t = t_i$ to $t = 0$
- 4: $\mathbf{p} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 5: **for** $j = 1, \dots, M$ **do**
- 6: $\epsilon_j \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 7: $(\hat{\mathbf{z}}_0, \mathbf{p}) = \text{Hamiltonian-Dynamics}(\hat{\mathbf{z}}_0, \mathbf{p}, \epsilon_j; \eta, \gamma)$ ▷ HMC updates for data consistency
- 8: **end for**
- 9: $\mathbf{z}_{t_{i-1}} \sim \mathcal{N}(\hat{\mathbf{z}}_0, \sigma_{t_{i-1}}^2 \mathbf{I})$ ▷ Proceed to the next noise level at time $t = t_{i-1}$
- 10: **end for**
- 11: **return** $\mathcal{D}(\mathbf{z}_{t_0})$ ▷ Return the decoded image

The algorithm’s main loop alternates between three key steps: (1) solving the PF-ODE backward from $t = t_i$ to $t = 0$ (line 3), (2) performing multi-step MCMC updates (lines 4–8), and (3) advancing to the next noise level (line 9). We will discuss each step in detail.

Solving PF-ODE backward from $t = t_i$ to $t = 0$ The probability flow ordinary differential equation (PF-ODE) [30] of the diffusion model, given by Equation (4), governs the continuous increase or reduction of noise in the image when moving forward or backward in time. Here, $\dot{\sigma}_t$ denotes the time derivative of σ_t , and $\nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t; \sigma_t)$ represents the time-dependent score function [47, 50].

$$d\mathbf{z}_t = -\dot{\sigma}_t \sigma_t \nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t; \sigma_t) dt, \quad (4)$$

Our goal is to solve the probability flow ODE (PF-ODE), as defined in Equation (4), backward from $t = t_i$ to $t = 0$, given the intermediate state $\mathbf{z}_{t_i}$ and the pretrained latent score function $\mathbf{s}_\theta(\mathbf{z}; \sigma) \approx \nabla_{\mathbf{z}} \log p(\mathbf{z}; \sigma)$. Any ODE solver, such as Euler’s method or the fourth-order Runge-Kutta method (RK4) [7], can be used to solve this problem. Following previous conventions [75], we adopt a few-step Euler method for solving it efficiently.

Multi-step MCMC updates Any MCMC samplers can be used, such as Langevin Dynamic Monte Carlo (LMC) and Hamiltonian Monte Carlo (HMC). For example, the LMC update with step size η is

$$\mathbf{z}_0^+ = \mathbf{z}_0 + \eta \nabla_{\mathbf{z}_0} \log p(\mathbf{y} \mid \mathcal{D}(\mathbf{z}_0)) + \eta \nabla_{\mathbf{z}_0} \log p(\mathbf{z}_0 \mid \mathbf{z}_t) + \sqrt{2\eta} \epsilon.$$

Note that the first gradient term can be computed with (2). The second gradient term, on the other hand, can be calculated by

$$\nabla_{\mathbf{z}_0} \log p(\mathbf{z}_0 \mid \mathbf{z}_t) = \nabla_{\mathbf{z}_0} \log p(\mathbf{z}_t \mid \mathbf{z}_0) + \nabla_{\mathbf{z}_0} \log p(\mathbf{z}_0) \approx \nabla_{\mathbf{z}_0} \log p(\mathbf{z}_t \mid \mathbf{z}_0) + \mathbf{s}_\theta(\mathbf{z}_0, t_{\min}).$$

This approximation holds for $t_{\min} \approx 0$, assuming that $\mathbf{z}_0$ lies close to the clean latent manifold [48]. To improve both convergence speed and approximation accuracy, the MCMC samplers are initialized with the solutions obtained from the previous PF-ODE step, leveraging its outputs as a warm start.

Note that during MCMC updates, the decoder $\mathcal{D}$ needs to be evaluated multiple times in the backward pass. To accelerate this process, we adopt Hamiltonian Monte Carlo (HMC), which typically requires fewer steps for convergence, thereby speeding up the algorithm. For each multi-step MCMC update, we introduce an additional momentum variable $\mathbf{p}$, initialized as $\mathcal{N}(\mathbf{0}, \mathbf{I})$. The Hamiltonian-Dynamics($\mathbf{z}_0, \mathbf{p}, \epsilon; \eta, \gamma$) update with step size η and damping factor γ is given by:

$$\mathbf{p}^+ = (1 - \gamma\eta) \cdot \mathbf{p} + \eta \nabla_{\mathbf{z}_0} \log p(\mathbf{z}_0 \mid \mathbf{z}_t) + \sqrt{2\gamma\eta} \epsilon \quad (5)$$

$$\mathbf{z}_0^+ = \mathbf{z}_0 + \eta \mathbf{p}^+ \quad (6)$$

Table 2: **Summary of STEP for black hole video reconstruction and dynamic MRI.** We provide and group the hyperparameters of Algorithm 1. HMC related parameters are tuned on 3 leave-out validation videos. The run time and memory are tested using 1 NVIDIA A 100-SCM4-80GB GPU.

	Black hole video reconstruction	Dynamic MRI
PF-ODE Related		
number of steps N_{ode}	20	20
scheduler σ_t	t	t
HMC Related		
number of steps M	60	53
scaling factor $1 - \gamma\eta$	0.00	0.83
step size square η^2	1.2e-5	1.2e-3
observation noise level σ_y	0.02	0.01
Decoupled Annealing Related		
number of steps N	25	20
final time T	100	100
discretization time $\{t_i\}, i = 1, \dots, N$	$\left(\frac{N-i}{N} \cdot T^{\frac{1}{7}}\right)^7$	$\left(\frac{N-i}{N} \cdot T^{\frac{1}{7}}\right)^7$
Inference Related		
decoder NFE N_{dec}	1500	1060
diffusion model NFE N_{dm}	500	400
time (s) per sample	645	332
memory (GB)	48	21

Proceeding to next noise level According to Proposition 1 in [75], one can obtain a sample $\mathbf{z}_{t_{i-1}} \sim p(\mathbf{z}_{t_{i-1}} \mid \mathbf{y})$ by simply adding Gaussian noise from a sample $\hat{\mathbf{z}}_0 \sim p(\mathbf{z}_0 \mid \mathbf{z}_{t_i}, \mathbf{y})$, given $\mathbf{z}_{t_i} \sim p(\mathbf{z}_{t_i} \mid \mathbf{y})$ from last step. Thus we solve the target posterior sampling by gradually sampling from the time-marginal posterior of diffusion trajectory. The full hyperparameters and running cost STEP is summarized in Tab. 2. The HMC-related parameters are searched on a leave out validation dataset consisting of 3 videos that are different from the testing videos.

B Experimental details

B.1 Black hole video reconstruction

We introduce the black hole video reconstruction problem in more detail. The goal is to reconstruct a video $\mathbf{x}_0 \in \mathbb{R}^{n_f \times n_h \times n_w}$ of a rapidly moving black hole. Each measurement, or *visibility*, is given by correlating the measurements from a pair of telescopes to sample a particular spatial Fourier frequency of the source with very long baseline interferometry (VLBI) [57, 74]. In VLBI, the cross-correlation of the recorded scalar electric fields at two telescopes, known as the ideal *visibility*, is related to the ideal source video $\mathbf{x}_0$ through a 2D Fourier transform, as given by the van Cittert-Zernike theorem [57, 74]. Specifically, the ideal visibility of the j -th frame of the target video is

$$\mathbf{I}_{\{a,b\}}^{[j]}(\mathbf{x}_0) := \int_{\rho} \int_{\delta} \exp\left(-i2\pi\left(u_{\{a,b\}}^{[j]}\rho + v_{\{a,b\}}^{[j]}\delta\right)\right) \mathbf{x}_0^{[j]}(\rho, \delta) d\rho d\delta \in \mathbb{C}, \quad (7)$$

where (ρ, δ) denotes the angular coordinates of the source video frame, and $(u_{\{a,b\}}^{[j]}, v_{\{a,b\}}^{[j]})$ is the dimensionless baseline vector between two telescopes $\{a, b\}$, orthogonal to the source direction.

Due to atmospheric turbulence and instrumental calibration errors, the observed visibility is corrupted by gain error, phase error, and additive Gaussian thermal noise [19, 53]:

$$\mathbf{V}_{\{a,b\}}^{[j]} := g_a^{[j]} g_b^{[j]} \exp\left(-i\left(\phi_a^{[j]} - \phi_b^{[j]}\right)\right) \mathbf{I}_{\{a,b\}}^{[j]}(\mathbf{x}_0) + \mathbf{n}_{\{a,b\}}^{[j]} \in \mathbb{C}. \quad (8)$$

where gain errors are denoted by $g_a^{[j]}, g_b^{[j]}$, phase errors are denoted by $\phi_a^{[j]}, \phi_b^{[j]}$, and thermal noise is denoted by $\mathbf{n}_{\{a,b\}}^{[j]}$. While the phase of the observed visibility cannot be directly used due to phase errors, the product of three visibilities among any combination of three telescopes, known as the *bispectrum*, can be computed to retain useful information. Specifically, the phase of the bispectrum, termed the *closure phase*, effectively cancels out the phase errors in the observed visibilities. Similarly, a strategy can be employed to cancel out amplitude gain errors and extract information from the visibility amplitude [3]. Formally, these quantities are defined as

$$\begin{aligned} \mathbf{y}_{\text{cp},\{a,b,c\}}^{[j]} &:= \angle(\mathbf{V}_{\{a,b\}}^{[j]} \mathbf{V}_{\{b,c\}}^{[j]} \mathbf{V}_{\{a,c\}}^{[j]}) \in \mathbb{R}, \\ \mathbf{y}_{\text{logca},\{a,b,c,d\}}^{[j]} &:= \log\left(\frac{|\mathbf{V}_{\{a,b\}}^{[j]}| |\mathbf{V}_{\{c,d\}}^{[j]}|}{|\mathbf{V}_{\{a,c\}}^{[j]}| |\mathbf{V}_{\{b,d\}}^{[j]}|}\right) \in \mathbb{R}. \end{aligned} \quad (9)$$

Here, $\angle(\cdot)$ denotes the complex angle, and $|\cdot|$ computes the complex amplitude. For a total of P telescopes, the number of closure phase measurements $\mathbf{y}_{\text{cp},\{a,b,c\}}^{[j]}$ at is $\frac{(P-1)(P-2)}{2}$, and the number of log closure amplitude measurements $\mathbf{y}_{\text{logca},\{a,b,c,d\}}^{[j]}$ is $\frac{P(P-3)}{2}$, after accounting for redundancy.

Let $D_{\text{cp}}^{[j]}$ and $D_{\text{camp}}^{[j]}$ to indicate the dimension of $\mathbf{y}_{\text{cp}}^{[j]}$ and $\mathbf{y}_{\text{logca}}^{[j]}$. Since closure quantities are nonlinear transformations of the visibilities, the black hole video reconstruction problem is non-convex.

To aggregate data over different measurement times and telescope combinations, the forward model of black hole video reconstruction for the j -th frame can be expressed as

$$\mathbf{y}^{[j]} := [\mathcal{A}_{\text{cp}}^{[j]}(\mathbf{x}_0), \mathcal{A}_{\text{logca}}^{[j]}(\mathbf{x}_0), \mathcal{A}_{\text{flux}}^{[j]}(\mathbf{x}_0)] := [\mathbf{y}_{\text{cp}}^{[j]}, \mathbf{y}_{\text{logca}}^{[j]}, \mathbf{y}_{\text{flux}}^{[j]}], \quad (10)$$

where $\mathbf{y}_{\text{cp}}^{[j]} = [\mathbf{y}_{\text{cp},\{a,b,c\}}^{[j]}]$ is the set of all closure phase measurements and $\mathbf{y}_{\text{cp}}^{[j]} = [\mathbf{y}_{\text{logca},\{a,b,c,d\}}^{[j]}]$ is the set of all log closure amplitude measurements for j -th frame. The total flux of the at j -th frame, representing the DC component of the Fourier transform, is given by

$$\mathbf{y}_{\text{flux}}^{[j]} := \int_{\rho} \int_{\delta} \mathbf{x}_0^{[j]}(\rho, \delta) d\rho d\delta. \quad (11)$$

The overall data consistency is an aggregation over all frames and typically expressed using the χ^2 statistics

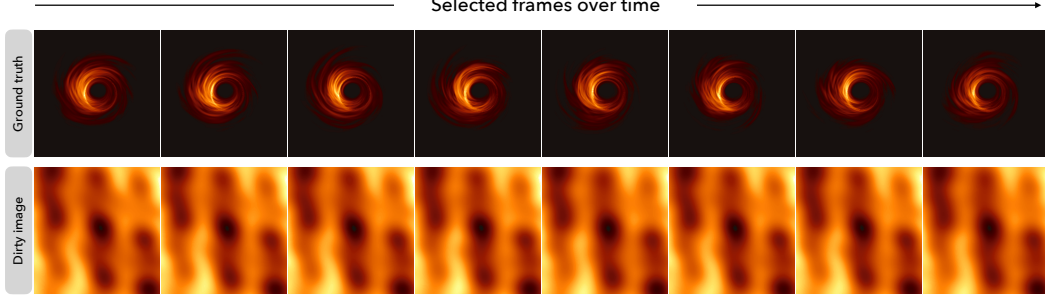


Figure 8: **The dirty images from the ideal visibilities.** We use the standard implementation in EHT library to get dirty images for each selected frame.

$$\begin{aligned}
-\log p(\mathbf{y} | \mathbf{x}_0) \propto & \underbrace{\sum_{j=1}^{n_f} \frac{1}{n_f D_{\text{cp}}^{[j]} \sigma_{\text{cp}}^2} \left\| \mathcal{A}_{\text{cp}}^{[j]}(\mathbf{x}_0) - \mathbf{y}_{\text{cp}}^{[j]} \right\|^2}_{\chi_{\text{cp}}^2} + \underbrace{\sum_{j=1}^{n_f} \frac{1}{n_f D_{\text{logca}}^{[j]} \sigma_{\text{logca}}^2} \left\| \mathcal{A}_{\text{logca}}^{[j]}(\mathbf{x}_0) - \mathbf{y}_{\text{logca}}^{[j]} \right\|^2}_{\chi_{\text{logca}}^2} \\
& + \beta \underbrace{\sum_{j=1}^{n_f} \frac{1}{n_f \sigma_{\text{flux}}^2} \left\| \mathcal{A}_{\text{flux}}^{[j]}(\mathbf{x}_0) - \mathbf{y}_{\text{flux}}^{[j]} \right\|^2}_{\chi_{\text{flux}}^2}
\end{aligned} \tag{12}$$

where σ_{cp} , σ_{logca} , and σ_{flux} are the estimated standard deviations of the measured closure phase, log closure amplitude, and flux, respectively, and β is a hyperparameter that controls the strength of the flux regularization, which is empirically determined. To evaluate the data fitting, we introduce a unified $\tilde{\chi}^2$ statistics

$$\tilde{\chi}^2 = \chi^2 \cdot \mathbb{1}\{\chi^2 \geq 1\} + \frac{1}{\chi^2} \cdot \mathbb{1}\{\chi^2 < 1\}. \tag{13}$$

The $\tilde{\chi}^2$ is no less than 1 and closer to 1, indicating better measurement data fit. In our experiments we use an average $(\tilde{\chi}_{\text{cp}}^2 + \tilde{\chi}_{\text{camp}}^2)/2$ for the data misfit metrics for evaluation.

Our experiments are based on the simulation of observing the Sagittarius A* black hole with the EHT 2017 array of eight radio telescopes over an observation period of around 100 minutes. We refer the readers to Fig. 5 of [37] for a visualization of the measurement patterns in Fourier space over time. To show the difficulty of this black hole video reconstruction problem, we visualize the dirty video frames obtained by applying the inverse Fourier transform to the ideal visibilities, assuming no measurement errors, in Figure 8. One can see that substantial spatiotemporal information is lost during the measurement process, so obtaining high-quality reconstructions relies on the effectiveness of incorporating prior information in the reconstruction process.

B.2 Dynamic MRI

MRI is an important imaging technique for clinical diagnosis and biomedical research, where the objective is to recover a video $\mathbf{x}_0 \in \mathbb{C}^{n_f \times n_h \times n_w}$ of the heart from the subsampled Fourier space (a.k.a k -space) measurements $\mathbf{y}$. Despite its many advantages, MRI is known to be slow because of the physical limitations of the data acquisition in k -space. This leads to low patient throughput and sensitivity to patient's motion [61]. To accelerate the scan speed, instead of fully sampling k -space, the compressed subsampling MRI (CS-MRI) technique subsamples k -space with masks $\{\mathbf{m}^{[j]}\}_{j=1}^{n_f}$. Mathematically, this can be formulated as

$$\mathbf{y}^{[j]} = \mathbf{m}^{[j]} \odot \mathcal{F}(\mathbf{x}_0^{[j]}) + \mathbf{n}^{[j]} \in \mathbb{C}^n \quad \text{for } j = 1, \dots, n_f, \tag{14}$$

where $\mathbf{m}^{[j]} \in \{0, 1\}^{n_h \times n_w}$ is the subsampling mask for the j -th frame, $\odot$ denotes element-wise multiplication, $\mathcal{F}$ is the Fourier transform, and $\mathbf{n}^{[j]}$ is the measurement noise. In our experiments, we

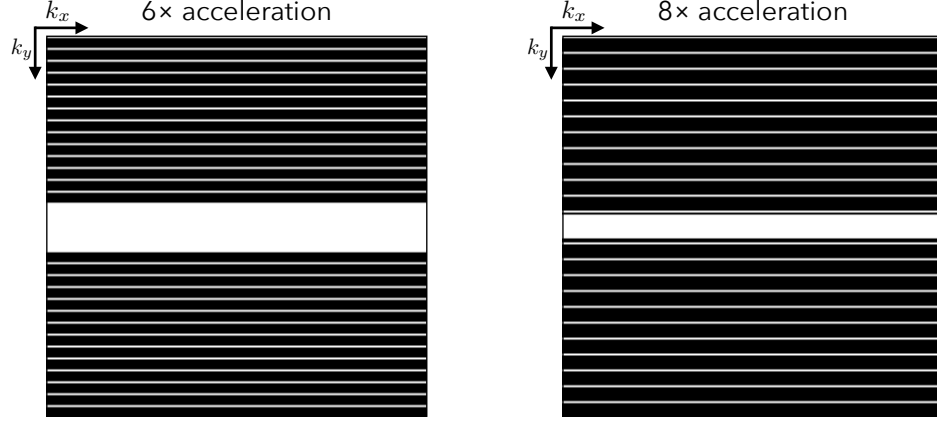


Figure 9: **Subsampling masks of $6\times$ (left) and $8\times$ (right) accelerations used in dynamic MRI experiments.** The white areas in the center indicate the auto-calibration (ACS) signals. The horizontal and vertical directions are the frequency (k_x) and phase (k_y) encoding directions, respectively. The same mask is applied to the sampling of each individual frame of all videos.

used subsampling masks with an equi-spaced pattern (similar to those visualized in [61]) of both $6\times$ acceleration with 24 auto-calibration signal (ACS) lines (Tab. 4) and $8\times$ acceleration with 12 ACS lines (Tab. 1). For dynamic MRI, we use the Gaussian likelihood function

$$-\log p(\mathbf{y}|\mathbf{x}_0) \propto \|\mathcal{A}(\mathbf{x}_0) - \mathbf{y}\|_2^2. \quad (15)$$

Fig.9 visualizes the subsampling masks used in our experiments, where k_x, k_y indicate the frequency encoding and phase encoding directions, respectively. The same mask is applied to the sampling of each frame of all videos.

B.3 Baseline implementations

In this section, we explain the detailed implementation of our baseline methods. We follow the same grouping in Sec. 4.1.

Group 1: simple heuristics We implement *batch independent sampling* (BIS) and *batch consistent sampling* (BCS) following [36, 35]. BIS is implemented to reconstruct each frame independently, whereas BCS promotes temporal consistency by using identical ("batch-consistent") noise across the temporal dimension. To ensure a fair comparison, we adapt Algorithm 1 by initializing the noise variables $\mathbf{z}_{t_N}$ (line 1) and $\mathbf{z}_{t_{i-1}}$ (line 9) with batch-consistent noise.

Group 2: noise warping We follow [16, 9] in performing noise warping based on optical flow estimated from measurements. As described in Sec. 4.1, we derive optical flow using ground-truth black hole videos and inverse Fourier-transformed dynamic MRI videos with a pretrained model [54]. Since we utilize a latent diffusion model, we further downsample the optical flow via interpolation to match the latent noise dimension. We adapt Algorithm 1 accordingly by replacing the original *i.i.d.* noise with the warped noise while keeping other components unchanged. We implement *Bilinear*, *Bicubic*, and *Nearest* warping strategies using their corresponding interpolation methods, and follow the implementation from https://github.com/yitongdeng-projects/infinite_resolution_integral_noise_warping_code for $\int$ -noise. Lastly, since [15] does not provide publicly available code, we implement GP-Warp following Equation (2) from their paper.

C Training details for spatiotemporal diffusion prior

In this section, we show the detail of getting a video diffusion prior on black hole video reconstruction and dynamic MRI, and we summarize the training hyper-parameters in Table 3. We define D_{image} and D_{video} as the image and video datasets, containing N_{image} and N_{video} data points, respectively. The image dataset D_{image} includes all individual frames from the video dataset D_{video} , along with additional large-scale image data to enhance generalization. For data augmentation, we apply random horizontal/vertical flipping and random zoom-in-and-out to improve robustness and diversity in training.

We first train the compression functions, the encoder $\mathcal{E}$ and decoder $\mathcal{D}$, on an image dataset. The training objective consists of an L1 reconstruction loss combined with a KL divergence term scaled by a factor β_{KL} . The loss function for training is as defined in Equation (16). The Adam optimizer is used as the default optimizer throughout the paper. The loss function for training the variational autoencoder (VAE) is given by:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_{\phi}(\mathbf{z}_0|\mathbf{x}_0), \mathbf{x}_0 \sim D_{\text{image}}} [\|\mathcal{D}(\mathbf{z}_0) - \mathbf{x}_0\|_1] + \beta_{\text{KL}} D_{\text{KL}}(q_{\phi}(\mathbf{z}_0|\mathbf{x}_0) \| p(\mathbf{z}_0)) \quad (16)$$

where $p(\mathbf{z}_0)$ is the standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and $q_{\phi}(\mathbf{z}_0|\mathbf{x}_0)$ is the isotropic Gaussian distribution over $\mathbf{z}_0$ where the mean and standard deviation is given by $\mathcal{E}(\mathbf{x}_0)$. Next, we train the image diffusion UNet $\mathbf{s}_{\theta}$ using the standard score-matching loss, as defined in Equation (17), following [26, 50].

$$\mathcal{L}_{\text{IDM}} = \mathbb{E}_{\mathbf{z}_0 \sim q_{\phi}(\mathbf{z}_0|\mathbf{x}_0), \mathbf{x}_0 \sim D_{\text{image}}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), t \sim \mathcal{U}(0, 1)} \left[\sigma_t^2 \|\mathbf{s}_{\theta}(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{x}_0)\|^2 \right] \quad (17)$$

After pertaining, the image diffusion UNet $\mathbf{s}_{\theta}$ is then converted to a spatiotemporal UNet by adding zero-initialized temporal modules to 2D spatial modules and fine-tune jointly with video and image

Table 3: **Summary of the training of spatiotemporal diffusion prior.** We provide and group the hyperparameters according to each component in the model. The model is trained with 1 NVIDIA A100-SCM4-80GB GPU.

Hyper-parameters	Black hole video reconstruction	Dynamic MRI
Dataset Related		
frames n_f	64	12
resolution $n_h \times n_w$	256×256	192×192
N_{image}	50000	39888
N_{video}	648	3324
VAE Training Related		
latent channels	1	2
block channels	[64, 128, 256, 256]	[256, 512, 512]
down sampling factor	8	4
batch size	16	16
epochs	25	10
β_{KL}	0.06	0.03
IDM Training Related		
block channels	[128, 256, 512, 512]	[128, 256, 512, 512]
batch size	16	16
epochs	200	50
Joint Fine-tuning Related		
p_{joint}	0.8	0.8
epochs	500	300
Other Info		
VAE parameters	14.8M	57.5M
diffusion model parameters	131.7M	131.7M
VAE training time	4.5h	8.9h
image diffusion model training time	5.5h	3.8h
joint fine-tuning time	13.7h	22.8h

datasets. We use the same Equation (17) without change $\mathbf{x}_0$ to video or pseudo video input and use the encoder to process each frame independently.

After pretraining, the image diffusion U-Net $\mathbf{s}_\theta$ is transformed into a spatiotemporal UNet by integrating zero-initialized temporal modules into the existing 2D spatial modules. The model is then fine-tuned jointly using both video and image datasets. We use the same loss as in Equation (17), by changing $\mathbf{x}_0$ to a video or a pseudo-video input. Each frame is independently processed using the encoder $\mathcal{E}$, ensuring that spatial representations remain aligned while temporal consistency is learned through the added temporal modules.

D Limitations

Though STEP is a general framework for solving scientific VIPs with spatiotemporal diffusion prior, the sampling cost of STEP is relatively high due to the requirement of backpropagation through decoder $\mathcal{D}$ in MCMC updates in Algorithm 1. So we have to strike a balance between the capability of the decoder and its computational cost. We leave the exploration of performing MCMC updates in pixel space or other better approaches to bypass backpropagation through the decoder as future work.

E Broader impacts

Our work introduces a new method for solving scientific video inverse problems, with potential applications in astronomy, medical imaging, and physics, where accurate reconstruction from sparse data is crucial. However, the learned spatiotemporal prior may introduce biases; thus, practitioners should responsibly evaluate the reconstructions and mitigate any potential negative impacts.

F Licenses

We list the licenses of all the assets we used in this paper:

- Data
 - GRMHD black hole data: Unknown
 - CMRxRecon Challenge 2023 [61]: MIT License
- Code
 - $\int$ -noise [16]²: MIT License
 - Diffusers [60]³: Apache-2.0 license
 - Common metrics on video quality⁴: None

² $\int$ -noise: https://github.com/yitongdeng-projects/infinite_resolution_integral_noise_warping_code

³Diffusers: <https://github.com/huggingface/diffusers>

⁴Common metrics on video quality: https://github.com/JunyaoHu/common_metrics_on_video_quality

Table 4: **Additional results on dynamic MRI with $6\times$ acceleration.** Following the same setup as in Tab. 1, we report the quantitative results on 10 test videos.

Tasks	Methods	PSNR ($\uparrow$)	SSIM ($\uparrow$)	LPIPS ($\downarrow$)	d-PSNR ($\uparrow$)	d-SSIM ($\uparrow$)	FVD ($\downarrow$)	Data Misfit ($\downarrow$)
MRI ($6\times$)	BIS [36]	39.47 (0.59)	0.958 (0.007)	0.086 (0.011)	43.26 (1.23)	0.962 (0.005)	113.17	11.071 (0.740)
	BCS [36]	40.69 (0.57)	0.959 (0.006)	0.081 (0.012)	44.73 (1.31)	0.974 (0.004)	110.01	11.085 (0.720)
	Bilinear [9]	40.85 (0.57)	0.960 (0.006)	0.080 (0.012)	44.84 (1.28)	0.975 (0.004)	114.37	11.038 (0.735)
	Bicubic [9]	40.71 (0.67)	0.959 (0.007)	0.079 (0.012)	44.74 (1.38)	0.974 (0.005)	106.82	11.068 (0.755)
	Nearest [9]	40.37 (0.56)	0.960 (0.007)	0.080 (0.012)	44.81 (1.41)	0.974 (0.005)	110.91	11.050 (0.739)
	λ -noise [9, 16]	40.09 (0.50)	0.960 (0.006)	0.082 (0.012)	44.77 (1.34)	0.974 (0.005)	111.92	11.059 (0.731)
	GP-Warp [15]	39.50 (0.48)	0.959 (0.007)	0.080 (0.012)	44.53 (1.33)	0.973 (0.005)	105.70	11.070 (0.727)
	STEP (video only)	40.76 (0.43)	0.967 (0.005)	0.077 (0.012)	46.38 (1.82)	0.981 (0.005)	101.83	10.788 (0.713)
	STEP (image-video joint)	41.39 (0.52)	0.969 (0.005)	0.076 (0.012)	46.61 (1.72)	0.982 (0.004)	98.15	<u>10.808</u> (0.723)

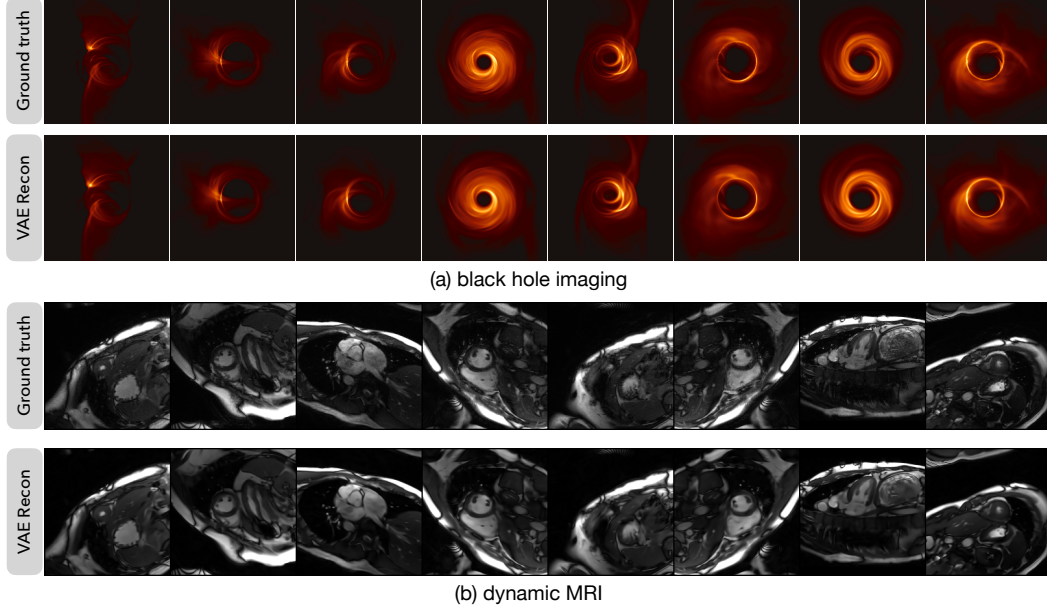


Figure 10: **Visualization of VAE reconstructions.** The VAE reconstructions are computed by first encoding the ground truth videos and then decoding them.

G More results & visualization

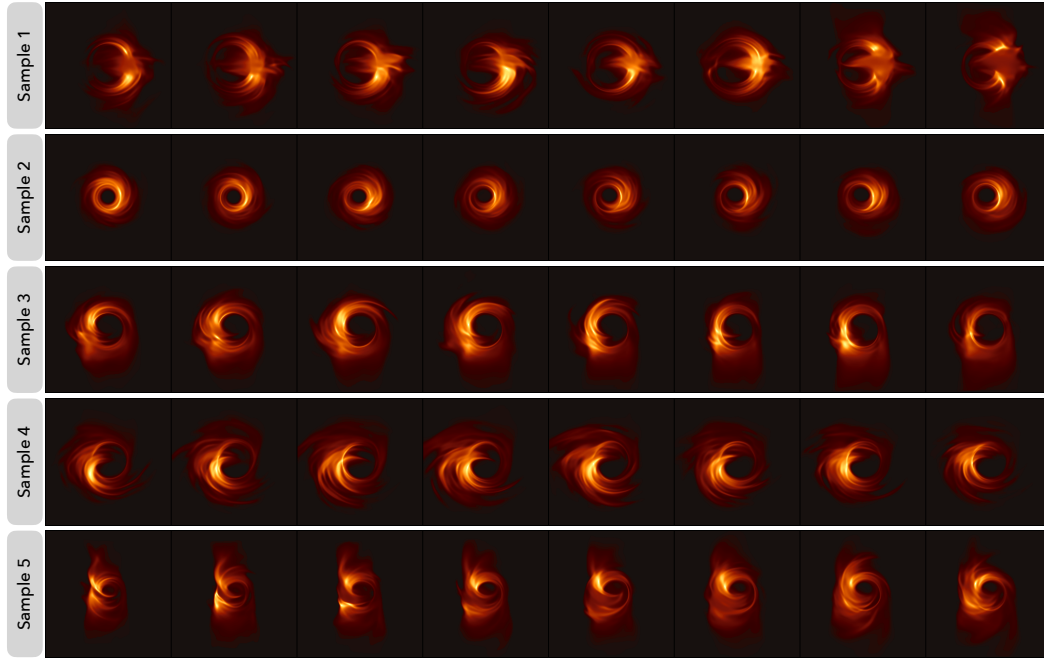
Data misfit values for samples in Fig. 1. We report the data misfit values in Tab. 5.

Table 5: **The data misfit values for samples shown in Fig. 1.** We report the data misfit metrics for the three obtained modes, which demonstrate that all modes fit the measurement data equally well.

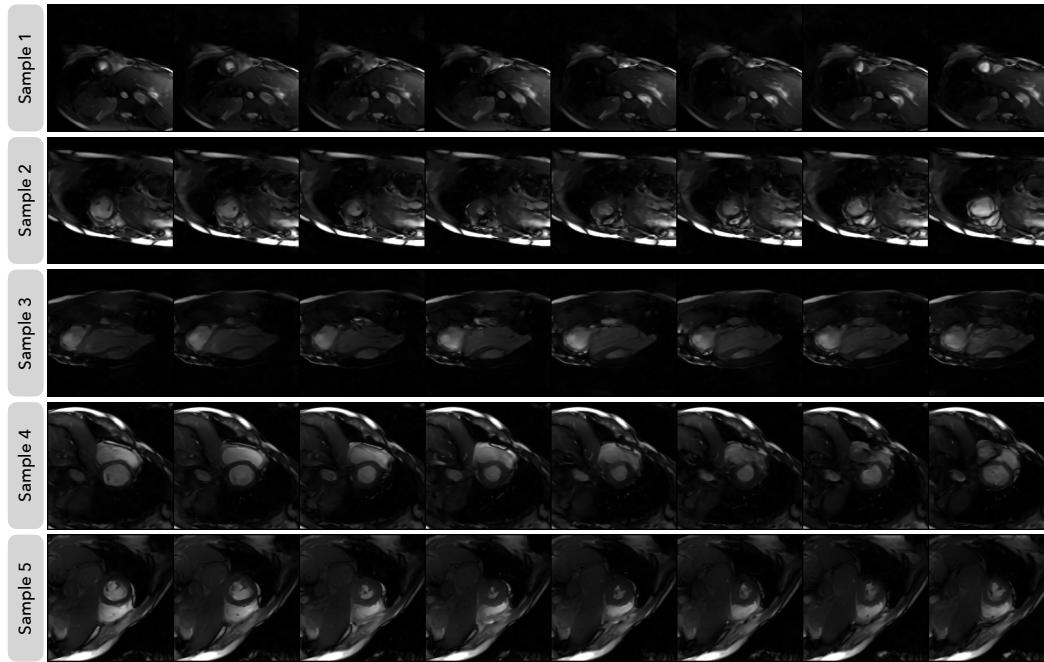
Metrics	Mode 1	Mode 2	Mode 3
χ_{cp}^2	1.045	0.987	1.084
χ_{logca}^2	1.007	1.001	1.202
Data Misfit	1.026	1.007	1.143

Dynamic MRI with higher acceleration. To access the capability of using spatiotemporal prior for solving more challenging inverse problems, we increase the acceleration times in Dynamic MRI, which makes the observation more sparse. The results are summarized in Table 4.

More visualizations Here, we show the VAE reconstruction results in Figure 10, unconditional samples in Figure 11 and additional posterior samples in Figure 12.



(a) black hole imaging



(b) dynamic MRI

Figure 11: **Visualization of video diffusion model unconditional samples.** The videos are sampled by solving PF-ODE with 100 Euler's steps.

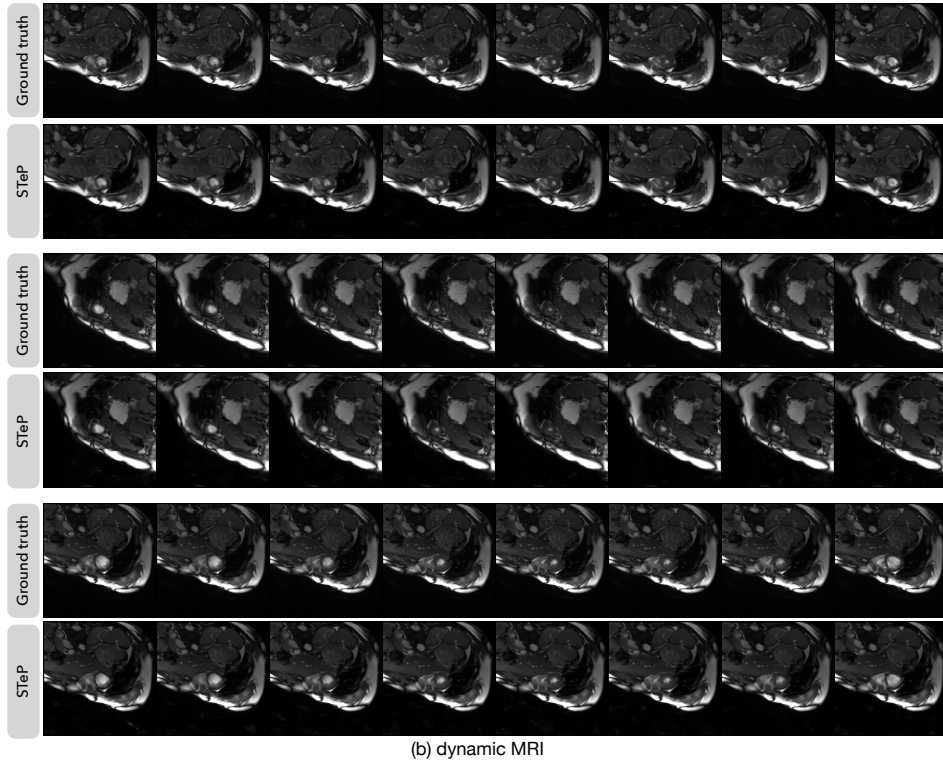
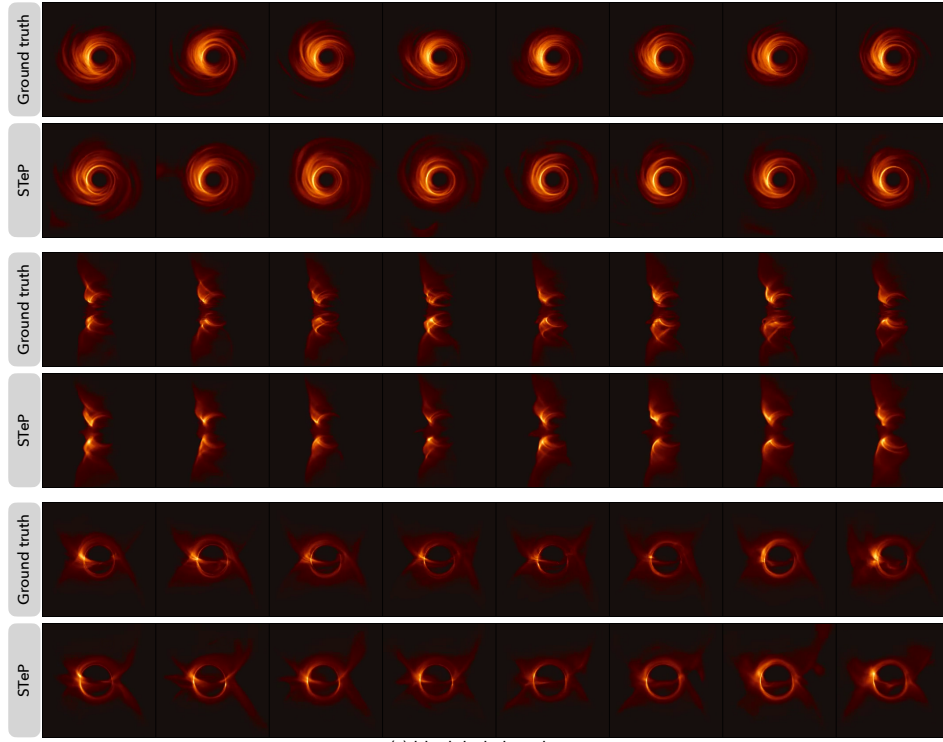


Figure 12: **Visualization of STEP posterior samples.** The videos are sampled using the Algorithm 1.